

1. Original tables

Account

Ano	Bname	Balance
101	JP Naga	10000
102	BTM	20000

Depositor

Dname	Ano
Sumit h	101
Ravi	102
Kiran	103

Goal written on slide:

“Find customer name having balance at least < 15000 ” – the English is inconsistent; the algebra and the red- crossing show that we actually want **balance < 15000** (i.e., less than 15000).

2. Cartesian product: Account $\times$ Depositor

First we form the **cross product** of Account and Depositor, pairing every account row with every depositor row.

Result (6 rows):

Acc.No	Bname	Balance	Dname	d.No
101	JP Naga	10000	Sumith	101
101	JP Naga	10000	Ravi	102
101	JP Naga	10000	Kiran	103
102	BTM	20000	Sumith	101
102	BTM	20000	Ravi	102
102	BTM	20000	Kiran	103

On the slide, all six combinations are shown; the red strikes indicate rows rejected by later conditions.

3. Selection step: $\sigma_{\{a.ano = D.ano \wedge bal < 15000\}}(\text{Account} \times \text{Depositor})$

We now apply the **selection** operator with two conditions:

- **Account.No = Depositor.No** (join condition: depositor's account number must match the account row).
- **Balance < 15000** (balance condition from the problem).

Apply these conditions row by row:

1. (101, JP Naga, 10000, Sumith, 101)
 - $101 = 101 \rightarrow$ matches.
 - $10000 < 15000 \rightarrow$ true.
 - ⇒ Row **kept** (boxed on slide).
2. (101, JP Naga, 10000, Ravi, 102)
 - $101 \neq 102 \rightarrow$ fails join condition.
 - ⇒ Row **discarded** (red line).
3. (101, JP Naga, 10000, Kiran, 103)

- $101 \neq 103 \rightarrow$ discard.
- 4. (102, BTM, 20000, Sumith, 101)
 - $102 \neq 101 \rightarrow$ discard.
- 5. (102, BTM, 20000, Ravi, 102)
 - $102 = 102$ (join ok) but $20000 < 15000$ is **false**.
 $\Rightarrow$ discard (purple/red line).
- 6. (102, BTM, 20000, Kiran, 103)
 - $102 \neq 103 \rightarrow$ discard.

After σ , only **one** tuple remains:

Acc. Ano	Bname	Balance	Dname	d. Ano
101	JP Naga	10000	Sumith	101

This is exactly what the boxed row on the slide represents.

Interpretation:

- Account 101 has balance 10000 (< 15000).
- Depositor Sumith is linked to account 101.
- No other depositor satisfies both “same account” and “balance < 15000 .”

4. Projection step: $\pi_{\{Dname\}}(\dots)$

Finally, we apply **projection** to keep only the depositor name from the filtered tuples.

$$\pi_{Dname}(\sigma_{a.ano=D.ano \wedge bal < 15000}(Account \times Depositor))$$

From the single remaining row, we project Dname:

Dname
Sumith

So the answer set (a relation) contains just one value: **{Sumith}**.

5. Intuitive reading of the expression

You can “read” the relational algebra left to right as:

1. Take all possible combinations of accounts and depositors: **Account** $\times$ **Depositor**.
2. From those, keep only pairs where the depositor’s account number matches the account and the balance is less than 15000: **$\sigma_{\{a.ano = D.ano \wedge bal < 15000\}}$** .
3. From the surviving rows, list only depositor names: **$\pi_{\{Dname\}}$** .

Result: names of customers whose linked account has balance < 15000, which in this data is **Sumith**.